

A rigorously characterized I_{3322} quantum wall, spatial attainment, and finite-dimensional nonattainment

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Abstract

We determine the supremum of the I_{3322} Bell functional in both the tensor-product and commuting-operator models. The value is the rigorously characterized domain-wall constant

$$q_* = 0.250875384513976536\dots, \quad q_* \in [0.250875384513976535514, 0.250875384513976536486].$$

A computer-assisted shooting argument produces a positive Bellman fixed point, an $\ell^2(\mathbb{Z})$ domain-wall eigenvector, and finite tensor-product strategies approaching q_* . Installing the wall in the alternating Pál–Vértesi projector blocks gives an explicit normal spatial strategy that attains q_* . A representation-free operator certificate, obtained by geometrically combining a Bellman inequality with its reflected counterpart, gives the matching commuting-operator upper bound. Equality on a finite joint spectrum forces two local decreasing transports to coincide. Their amplitude compatibility then bounds every putative finite attaining component by $1/4$. Since $q_* > 1/4$, no finite-dimensional tensor-product or commuting-operator strategy attains the supremum. Thus the maximizing correlation lies in $C_{qs}(3, 3; 2, 2) \setminus C_q(3, 3; 2, 2)$. The displayed decimal is a validated shooting zero, not an asserted elementary closed form. As a corollary, the finite-dimensional quantum correlation set in the three-input binary scenario is not closed; this is the minimal bipartite binary-output scenario by input counts where nonclosure and finite/spatial separation can occur. The wall's principal sections also obey an exact boundary-flux law. If Q_d allows binary POVMs on local dimension at most d , then

$$\kappa d^{-4} \Gamma^{-d} \leq q_* - Q_d \leq \exp[-d \log R + O(1)],$$

where $R = 1.07809205080209208\dots$, $\Gamma = 312^4 = 9,475,854,336$, and one explicit admissible κ is approximately $4.294654614331446 \times 10^{-52}$. The upper estimate is attained by explicit truncations and the lower estimate holds for every strategy in the stated dimension class. Hence the minimum local dimension for deficit ε is $\Theta(\log(1/\varepsilon))$; the constants are not claimed sharp.

1 Statement of the result

Let A_1, A_2, A_3 and B_1, B_2, B_3 be projections. In the commuting model they act on one Hilbert space and $[A_i, B_j] = 0$. We use the normalization

$$\begin{aligned} \mathcal{B}_{3322} = & -A_2 - B_1 - 2B_2 + A_1B_1 + A_1B_2 + A_2B_1 + A_2B_2 \\ & - A_1B_3 + A_2B_3 - A_3B_1 + A_3B_2. \end{aligned} \tag{1}$$

This is the inequality introduced by Collins and Gisin [4]. Its classical bound is zero. Qubit strategies attain $1/4$. Pál and Vértesi constructed increasing-dimensional strategies approaching approximately 0.250875384514 and conjectured finite-dimensional nonattainment [11]. Our projector convention agrees

term-for-term with their best-response recursion. In the standard Collins–Gisin table it is obtained by swapping the first two settings on both sides; the exact affine conversion to the dichotomic convention is recorded in the accompanying normalization concordance.

Theorem 1. *There is a rigorously characterized constant q_* satisfying*

$$q_* \in [0.250875384513976535514, 0.250875384513976536486] \quad (2)$$

for which

$$\omega_{\otimes}(\mathcal{B}_{3322}) = \omega_c(\mathcal{B}_{3322}) = q_*.$$

No finite-dimensional tensor-product strategy and no finite-dimensional commuting-operator strategy attains q_ . A normal vector state on $\ell^2(\mathbb{Z}) \otimes \ell^2(\mathbb{Z})$ does attain q_* .*

The NPA hierarchy [10] and its recent ncKKT refinements [1] provide increasingly sharp numerical bounds for this problem. Our proof instead extracts an exact operator certificate from the limiting domain-wall recursion. Infinite-versus-finite dimensional separations and non-closure are known in larger Bell scenarios [13, 3]; the point here is the exact resolution of the canonical three-setting binary functional. Mghirbi previously gave proof-carrying exact rational upper and lower certificates enclosing the value to width below 10^{-9} [8] ([doi:10.5281/zenodo.21477901](https://doi.org/10.5281/zenodo.21477901)). That important antecedent explicitly left the exact supremum, finite-dimensional attainment, and tensor–commuting equality open. Very recent budgeted moment-selection work continues to use I_{3322} as a hard NPA benchmark rather than supplying an exact theorem [6].

1.1 Minimal binary spatial separation and nonclosure

Let $C_q(m_A, m_B; 2, 2)$ denote the set of finite-dimensional bipartite quantum correlations with m_A and m_B binary measurements. We allow POVMs; this is equivalent to the finite-dimensional PVM convention. Let C_{qs} denote the same correlations with arbitrary spatial tensor-product Hilbert spaces and normal states.

Theorem 2 (Minimal binary spatial separation). *There is a correlation*

$$p_* \in C_{qs}(3, 3; 2, 2) \setminus C_q(3, 3; 2, 2) \quad \text{with} \quad I_{3322}(p_*) = q_*.$$

Consequently $C_q(3, 3; 2, 2)$ is not closed. If either $m_A \leq 2$ or $m_B \leq 2$, then $C_q = C_{qs}$ and the common set is compact. Hence $(3, 3; 2, 2)$ is minimal, under the coordinatewise ordering of input counts, for both phenomena among bipartite binary-output scenarios.

Proof. Section 3 constructs the asserted p_* explicitly. Theorem 1 excludes every finite-dimensional attainer, so $p_* \notin C_q$. Since $C_{qs} \subseteq \overline{C_q}$, this also proves nonclosure. The implication from I_{3322} nonattainment to this statement was explicitly anticipated by Dykema, Paulsen, and Prakash [5]; general $C_q \neq C_{qs}$ separations were proved by Coladangelo and Stark [3].

For minimality, consider $p \in C_q(2, m; 2, 2)$. Simultaneously dilate Alice’s two binary POVMs to two projections. Jordan’s lemma decomposes their common finite-dimensional space into invariant blocks of dimension at most two. Alice’s block-diagonal measurements make p a convex mixture of the corresponding block correlations. After purifying each block state into Bob’s system, Schmidt compression reduces Bob’s support to dimension at most two without changing any probability. Thus

$$C_q(2, m; 2, 2) = \text{conv}(K_m),$$

where K_m is the compact set of qubit–qubit behaviors with the prescribed measurements. The convex hull of a compact subset of a finite-dimensional space is compact. The standard inclusions

$C_q \subseteq C_{qs} \subseteq \overline{C_q}$ then force $C_q = C_{qs}$. More explicitly, the no-signaling affine dimension is $3m + 2$, so Carathéodory reduces every behavior to at most $3m + 3$ qubit blocks, giving local dimensions at most $6m + 6$. The argument with the parties interchanged proves the other case. Related recent dimension bounds appear in [7, 12]. $\square$

The theorem is only a binary-output, bipartite setting-count statement. It does not compare scenarios with more outputs, more parties, communication, or network causal structure.

2 Certified Bellman datum

Put

$$b(x) = \frac{\sqrt{1-x^2}}{2}, \quad d(x, u) = xu + \frac{x-u}{2} - 1. \quad (3)$$

The only computer-assisted input to the analytic proof is the following.

Proposition 3 (Certified Bellman datum). *There exist q_* satisfying (2), a positive continuous concave function $F : [-1, 1] \rightarrow (0, \infty)$, and a strictly increasing map $P : [-1, 1] \rightarrow [-1, 1]$ such that, with*

$$p(x) = \frac{b(x)^2}{F(x)},$$

one has

$$p(x) + F(u) \leq q_* - d(x, u) \quad (4)$$

for every $x, u \in [-1, 1]$. For each u , equality holds at the unique point $x = P(u)$. Moreover, a sequence of finite-dimensional tensor-product strategies has Bell values converging to q_* . There also exist a profile $(c_j)_{j \in \mathbb{Z}} \subset (-1, 1)$ and a positive normalized vector $\lambda = (\lambda_j)_{j \in \mathbb{Z}} \in \ell^2(\mathbb{Z})$, with geometric tails, such that

$$H(c)\lambda = q_*\lambda, \quad H_{jj} = d(c_j, c_{j+1}), \quad H_{j-1,j} = b(c_j). \quad (5)$$

Equivalently, F is a fixed point of

$$(T_q F)(u) = \min_{x \in [-1, 1]} \left(q + 1 - \frac{x}{2} - (x - \frac{1}{2})u - \frac{1-x^2}{4F(x)} \right). \quad (6)$$

The validated construction of Proposition 3 is summarized in Section 8.

3 Normal spatial attainment

We now install the certified wall in the alternating block construction of Pál and Vértesi [11]. For $t \in [-1, 1]$, set $s_t = \sqrt{1-t^2}$ and

$$P_A^\pm(t) = \frac{1}{2} \begin{pmatrix} 1-t & \pm s_t \\ \pm s_t & 1+t \end{pmatrix}, \quad P_B^\pm(t) = \frac{1}{2} \begin{pmatrix} 1+t & \pm s_t \\ \pm s_t & 1-t \end{pmatrix}, \quad R = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}. \quad (7)$$

These are rank-one orthogonal projections. On each local copy of $\ell^2(\mathbb{Z})$, partition the standard basis into the two perfect matchings

$$E_A = \{(2k-1, 2k) : k \in \mathbb{Z}\}, \quad E_B = \{(2k, 2k+1) : k \in \mathbb{Z}\}.$$

On every edge $(j-1, j) \in E_A$, install

$$A_1 = P_A^-(c_j), \quad A_2 = P_A^+(c_j), \quad B_3 = R,$$

and on every edge $(j-1, j) \in E_B$, install

$$B_1 = P_B^-(c_j), \quad B_2 = P_B^+(c_j), \quad A_3 = R.$$

Each measurement is therefore an orthogonal direct sum of rank-one projections and defines a bounded projection on $\ell^2(\mathbb{Z})$.

Define the unit vector

$$\psi_* = \sum_{j \in \mathbb{Z}} \lambda_j e_j \otimes e_j \in \ell^2(\mathbb{Z}) \otimes \ell^2(\mathbb{Z}). \quad (8)$$

This gives a normal vector state. For the finite-band real matrices M, N used here,

$$\langle \psi_*, M \otimes N \psi_* \rangle = \sum_{r,s} \lambda_r \lambda_s M_{rs} N_{rs}.$$

In the present two-banded case all diagonal sums are dominated by $\sum_j \lambda_j^2$, while the neighbor sums converge absolutely by Cauchy–Schwarz. Direct expansion of (1) and (7) therefore gives

$$\begin{aligned} \langle \psi_*, \mathcal{B}_{3322} \psi_* \rangle &= \sum_j d(c_j, c_{j+1}) \lambda_j^2 + \sum_j \sqrt{1 - c_j^2} \lambda_{j-1} \lambda_j \\ &= \langle \lambda, H(c) \lambda \rangle = q_*. \end{aligned} \quad (9)$$

The first identity is independently guarded both on exact-rational open and endpoint-free fixtures and by a symbolic periodic reconstruction. The last identity is (5). Thus the correlation belongs to C_{qs} and attains the certified wall; finite-dimensional nonattainment places it outside C_q .

4 Quantitative finite-dimensional dimension law

Let Q_d be the supremum of $\mathcal{B}_{3322}$ over tensor-product strategies whose two local Hilbert spaces have dimension at most d , allowing binary POVMs. For later use, let $D(\varepsilon)$ be the least d for which $q_* - Q_d \leq \varepsilon$.

Dimension-constrained noncommutative hierarchies provide convergent numerical bounds at fixed local dimension, including applications to I3322 [9]. Quantitative dimension blowups substantially stronger than the one below are known for purpose-built nonlocal games with larger question or answer sets [2]. Our statement has a different scope: it gives the asymptotic order, from both sides, for the canonical three-setting binary functional itself.

Theorem 4 (Dimension law). *There is an explicit universal constant $\kappa > 0$ such that, for every $d \geq 1$,*

$$\kappa d^{-4} \Gamma^{-d} \leq q_* - Q_d \leq \exp[-d \log R + O(1)], \quad \Gamma = 312^4 = 9,475,854,336, \quad (10)$$

where R is given in (13). One admissible exact κ is defined in the supplement and satisfies

$$\kappa = 4.2946546143314459987 \dots \times 10^{-52}.$$

Consequently,

$$\frac{\log(1/\varepsilon) - O(\log \log(1/\varepsilon))}{\log \Gamma} \leq D(\varepsilon) \leq \frac{\log(1/\varepsilon)}{\log R} + O(1), \quad (11)$$

and, as $\varepsilon \downarrow 0$, $D(\varepsilon) = \Theta(\log(1/\varepsilon))$.

For the constructive half, for $L \geq 0$, put $I_L = \{-L, \dots, L\}$, $S_L = \sum_{j \in I_L} \lambda_j^2$, and $d = 2L + 1$. Compress the six spatial measurements from Section 3 to $\text{span}\{e_j : j \in I_L\}$. A compressed projection is a positive contraction, hence a valid binary effect. Let v_L be the Bell value of the normalized truncated state and write $h_j = \sqrt{1 - c_j^2}/2$.

Summing the wall eigenvalue equation over the principal section gives the exact identity

$$q_* - v_L = \frac{h_{-L}\lambda_{-L-1}\lambda_{-L} + h_{L+1}\lambda_L\lambda_{L+1}}{S_L}. \quad (12)$$

Every diagonal and internal nearest-neighbor term occurs in the principal Jacobi quadratic form; the two displayed edges are the only omitted terms. Positivity of the wall makes the deficit strictly positive.

The certified analytic tails satisfy

$$\frac{\lambda_{j+1}}{\lambda_j} \longrightarrow R^{-1} \quad (j \rightarrow +\infty), \quad \frac{\lambda_{j-1}}{\lambda_j} \longrightarrow R^{-1} \quad (j \rightarrow -\infty),$$

where

$$R = 1.07809205080209208\dots, \quad \log R = 0.07519285919570202\dots \quad (13)$$

The boundary coefficients tend to positive constants and $S_L \rightarrow 1$, hence

$$\lim_{L \rightarrow \infty} \frac{-\log(q_* - v_L)}{2L + 1} = \log R. \quad (14)$$

Consequently, along odd dimensions,

$$0 < q_* - Q_d \leq q_* - v_{(d-1)/2} = \exp[-d \log R + O(1)], \quad (15)$$

and even dimensions follow by padding with an unused summand. Thus deficit at most ε is achievable with

$$d \leq \frac{\log(1/\varepsilon)}{\log R} + O(1), \quad \frac{1}{\log R} = 13.2991351931\dots \quad (16)$$

The flux identity and plateau exponent have separate production and independent symbolic/interval reconstructions.

For the converse half, compactness and multilinear extremality first replace an optimizer by a pure state and six projective effects on the same local spaces; no dilation is used. Writing the Bell deficit as the sum of the three positive certificate remainders, robust contact coercivity localizes their joint spectral mass. A shifted moving grid follows canonical source–target packets through the two response maps. Near-fixed, inactive-tail, contact, and stopped-frame losses are each charged with their full reuse multiplicity. On the retained drift sector every packet has one predecessor and at most one successor, while each individual chain visits disjoint spectral projections of one local d -dimensional operator and hence has length at most d .

The resulting terminal-plus-recurrence energy satisfies both

$$E_{\text{exit}} + E_{\text{rec}} \leq Ad\varepsilon + BC_F 20^{2d} \sqrt{\varepsilon}$$

and

$$E_{\text{exit}} + E_{\text{rec}} \geq \frac{W_D}{(d+1)(78/5)^{2d}}, \quad W_D \geq 1 - C_I 20^{2d} \sqrt{\varepsilon},$$

with explicit dimension-independent constants $A = 5616$, $B = 200772/25$, C_F , and C_I . Splitting according to whether the retained mass is at least $1/2$, and then according to which upper-energy term is large, gives

$$\varepsilon \geq \kappa d^{-4} (20 \cdot 78/5)^{-4d},$$

which is the lower half of (10). The supplement gives the complete constant ledger and prohibited-shortcut audit. A sealed-source blind reconstruction independently reproduced Γ , A , B , and κ , followed by a separate exact SymPy audit.

5 Geometric symmetrization

Apply (4) at (x, u) and $(-u, -x)$. Since $d(-u, -x) = d(x, u)$,

$$\begin{aligned} p(x) + F(u) &\leq q_* - d(x, u), \\ F(-x) + p(-u) &\leq q_* - d(x, u). \end{aligned} \quad (17)$$

Define

$$a(x) = \sqrt{p(x)F(-x)}, \quad c(u) = \sqrt{F(u)p(-u)}. \quad (18)$$

Cauchy-Schwarz applied to $(\sqrt{p(x)}, \sqrt{F(-x)})$ and $(\sqrt{F(u)}, \sqrt{p(-u)})$ gives

$$a(x) + c(u) \leq q_* - d(x, u). \quad (19)$$

Direct multiplication yields

$$a(x)a(-x) = b(x)^2, \quad c(u)c(-u) = b(u)^2. \quad (20)$$

6 Commuting-operator certificate

Set

$$X = A_1 + A_2 - I, \quad Y = A_2 - A_1, \quad U = B_1 + B_2 - I, \quad V = B_2 - B_1. \quad (21)$$

The projection relations imply

$$X^2 + Y^2 = I, \quad XY + YX = 0, \quad U^2 + V^2 = I, \quad UV + VU = 0. \quad (22)$$

A direct expansion gives

$$\mathcal{B}_{3322} = G(X, U) + Y(B_3 - I/2) + (A_3 - I/2)V, \quad (23)$$

where

$$G(X, U) = XU + X/2 - U/2 - I.$$

Let

$$\alpha(x) = \frac{1}{2} - a(x), \quad \beta(u) = q_* - \frac{1}{2} - c(u).$$

Because X and U commute, joint functional calculus applied to (19) gives

$$R_0 := \alpha(X) + \beta(U) - G(X, U) \geq 0. \quad (24)$$

Let J_A be the polar sign of Y on the support of $I - X^2$. Then

$$Y = 2b(X)J_A, \quad J_A f(X) = f(-X)J_A. \quad (25)$$

Put $S_B = 2B_3 - I$ and $L_A = \sqrt{a(X)}$. Cross-party commutation gives $[J_A, S_B] = 0$. Using (20),

$$L_A J_A S_B L_A = b(X)J_A S_B = Y(B_3 - I/2). \quad (26)$$

Consequently

$$\begin{aligned} R_A &:= \frac{1}{2}I - \alpha(X) - Y(B_3 - I/2) \\ &= L_A(I - J_A S_B)L_A \geq 0. \end{aligned} \quad (27)$$

The product $J_A S_B$ is a self-adjoint contraction. Endpoint summands, where $b(X) = Y = 0$, follow by continuity. The identical Bob construction gives

$$R_B := (q_* - \tfrac{1}{2})I - \beta(U) - (A_3 - I/2)V = \sqrt{c(U)}(I - S_A J_B)\sqrt{c(U)} \geq 0. \quad (28)$$

Adding (24), (27), and (28) yields

$$\boxed{q_* I - \mathcal{B}_{3322} = R_0 + R_A + R_B \geq 0.} \quad (29)$$

Thus $\omega_c \leq q_*$. Proposition 3 supplies a tensor-product sequence approaching q_* , proving equality of both suprema.

7 Finite-dimensional nonattainment

Suppose a finite-dimensional strategy attains q_* , and let ψ be an attaining unit vector. Positivity in (29) implies

$$R_0 \psi = R_A \psi = R_B \psi = 0. \quad (30)$$

The scalar value of R_0 at a joint spectral pair (x, u) is $q_* - d(x, u) - a(x) - c(u)$. Equality forces equality in both inequalities of (17) and in Cauchy–Schwarz. Hence every occupied pair obeys

$$x = P(u), \quad -u = P(-x). \quad (31)$$

Let Σ be the finite set of occupied u -coordinates. Strict monotonicity makes P injective.

On the occupied interior support define

$$K_A = J_A S_B, \quad K_B = S_A J_B, \quad r_A(x) = \frac{a(x)}{b(x)}, \quad r_B(u) = \frac{c(u)}{b(u)}.$$

The local equations in (30) become

$$K_A \psi = r_A(X) \psi, \quad K_B \psi = r_B(U) \psi. \quad (32)$$

The first reverses x and therefore acts on Σ by

$$\mathfrak{a}(u) = P^{-1}(-P(u)), \quad (33)$$

while the second acts by

$$\mathfrak{b}(u) = -u. \quad (34)$$

Both are decreasing bijections of the same finite ordered set. Such a set has only one decreasing bijection, so $\mathfrak{a} = \mathfrak{b}$. Every occupied (x, u) is paired with $(-x, -u)$.

Let ψ_+ and ψ_- be the corresponding nonzero spectral components. Taking norms in (32) and using (20) gives

$$\rho := \frac{\|\psi_-\|}{\|\psi_+\|} = \sqrt{\frac{F(-x)}{F(x)}} = \sqrt{\frac{F(u)}{F(-u)}}. \quad (35)$$

Bellman and Cauchy equality then force

$$F(x) = \frac{b_x}{\rho}, \quad F(-x) = \rho b_x, \quad F(u) = \rho b_u, \quad F(-u) = \frac{b_u}{\rho}. \quad (36)$$

At the two reflected components,

$$q_* - d(x, u) = \rho(b_x + b_u), \quad q_* - d(-x, -u) = \frac{b_x + b_u}{\rho}. \quad (37)$$

Subtracting and eliminating ρ gives

$$q_* = xu - 1 + \sqrt{(b_x + b_u)^2 + \frac{(x - u)^2}{4}}. \quad (38)$$

Put $s_x = \sqrt{1 - x^2}$, $s_u = \sqrt{1 - u^2}$, and $t = 1 - xu \geq 0$. Then

$$(b_x + b_u)^2 + \frac{(x - u)^2}{4} = \frac{1 - xu + s_x s_u}{2} \leq t, \quad (39)$$

since

$$(1 - xu)^2 - s_x^2 s_u^2 = (x - u)^2 \geq 0.$$

Therefore

$$q_* \leq -t + \sqrt{t} \leq \frac{1}{4}, \quad (40)$$

contradicting (2). This proves finite-dimensional nonattainment. Independently, a finite-dimensional commuting representation decomposes into finitely many spatial matrix blocks; equality would force one finite tensor block to attain q_* .

8 Validated construction of the Bellman datum

The Bellman equality and stationarity equations close as a three-dimensional shooting map. In coordinates (z, x, r) , put

$$\Delta = zx + \frac{z - x}{2} - 1, \quad v = \frac{2(q - \Delta - \sqrt{1 - z^2}/(2r))}{\sqrt{1 - x^2}}.$$

The successor coordinate is the explicit rational-radical expression obtained from stationarity. The selected orbit leaves a hyperbolic positive plateau, meets a two-condition reflection section, and returns by reversing symmetry.

The validation stack comprises: a 400-bit Miranda enclosure of the shooting zero; a symbolic envelope-covariance identity; 300-bit interval covers of the central graph and endpoint wings; strict derivative signs on every tile; exclusion of all inactive outer predecessors; positivity of the Bellman function; and square-summable domain-wall tails whose finite truncations approach q_* . Every computer-assisted item was also reconstructed with a separate `mpmath.iv` implementation and locally written complex intervals, without importing the Arb/FLINT production modules. All gates pass and its enclosure overlaps the production result. The spatial block identity has two further independent guards: exact rational fixtures, including endpoint-free controls, and a symbolic polynomial reconstruction. The certificate map and technical supplement give every script, margin, failed guard, and the exact independence contract.

9 Scope

Theorem 1 concerns one Bell functional. It does not assert an elementary closed form for q_* , spatiality of arbitrary commuting correlations, uniqueness or self-testing of the maximizer, or experimental accessibility of the limiting wall. It separates C_q from C_{qs} at this correlation but does not separate C_{qs} from C_{qa} or C_{qa} from C_{qc} . The proof is independent of any broader foundational interpretation. The observed finite-carrier gaps are exponentially controlled above and below, but the optimal exponential necessity rate and prefactor remain unknown. The lower bound is device-independent only within the standard bipartite tensor-product model with no communication and both local dimensions bounded by d ; it does not cover signaling, postselection, network resources, or alternative notions of operator-algebraic dimension.

Computational disclosure. Frontier language models were used extensively for proof discovery, implementation, adversarial auditing, and editorial assistance. The named author assumes responsibility for the release. Every load-bearing computer-assisted claim is mapped to independently replayable certificates.

Artifacts. The complete sources, frozen receipts, and independent reconstruction are at <https://github.com/Apsiape/i3322-exact-wall>.

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